

DDTA - Scheda di lettura compilata

SRC-0039 - BA0-R systems-modeling prior art

REVIEW-ONLY: da approvare prima del caricamento nel repository.

Identita bibliografica e accesso	
Source ID	SRC-0039
Titolo	NASA Systems Modeling Handbook for Systems Engineering
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1. Research problem and contribution

Integrates system modeling with selected NASA systems-engineering processes and work products. The handbook is tool-agnostic and states that, although it uses SysML v1.7 examples, other modeling languages may be applied to its metamodel. It separates modeling language, methodology and framework and uses a metamodel to relate elements across structure, behavior, requirements and parametrics.

2. Starting artifacts and generated representations

The model captures stakeholder expectations, requirements, structure, behavior, interfaces, item/information flows, measures, verification and validation knowledge. System context captures scope/boundaries, the system of interest, users and external systems. Internal block views show structural interfaces and items flowing across them. Diagrams and tables are generated/extracted from the populated model rather than treated as the canonical knowledge store.

3. Traceability, change, automation and human role

The handbook treats model planning, conventions, accessibility, roles/responsibilities and model management as explicit planning concerns. It uses traceability relationships across requirements, measures, verification and structure/behavior elements. Rendering choices may omit ports or details for visual simplicity without deleting model content. Tooling supports extraction of views but does not replace modeling methodology or governance.

4. Findings, limitations and relationship with DDTA

Strong prior art for a methodology-neutral canonical system model with multiple projections. It directly weakens a minimal Actor+Asset+Interaction core because interface, flow, item, structure, behavior and mode are distinct responsibilities. It does not establish that NASA classes should become DDTA BAE metaclasses; its metamodel is tailored to NASA SE work products and SysML practice. It strongly supports Base Analysis as a canonical knowledge representation from which views are derived.

Evidenza utile per BA0/BA1

Pag.	Ruolo	Sezione	Interpretazione DDTA
5	foundation	1.1 Purpose	A modeling metamodel can be conceptually separated from its concrete notation/tool.
11	foundation	4.3.1.2 Modeling Pillars of SysML	Structure and behavior are distinct system-modeling responsibilities rather than one generic interaction bucket.
14	foundation	4.3.3 Modeling Framework	Canonical model content should precede view generation.

Pag.	Ruolo	Sezione	Interpretazione DDTA
23	foundation	8.3 System Context bdd	Boundary/context is a generic systems-modeling responsibility, not inherently a security trust boundary.
24	challenge	8.4 System Context ibd	Interface and transferred item/flow should not be collapsed into a single generic interaction.
45	foundation	9 Generating Diagrams and Tables	Views are projections/renderings of canonical knowledge, not independent sources of truth.
70	comparison	Appendix F 3.3 Interfaces	An interface abstraction can be broader than a communication channel.
72	foundation	Appendix F 3.4 Modes of Operations	State/mode is a recurring semantic responsibility that BA1 must test.

Bibliography mining / follow-up

- OMG SysML v1.7 and v2 specifications
- Friedenthal, Moore & Steiner, Practical Guide to SysML
- NASA OpenSE Cookbook / MBSE Grid references

Governance note

Questa scheda è un ausilio di review. Le future citazioni di tesi devono risolversi alla fonte registrata e a una posizione esatta nell'excerpt ledger approvato. Nessun concetto esterno diventa automaticamente un BAE DDTA.